Day 6 — HotpotQA Supporting-Fact Baseline Experiments

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Objective

Following completion of the T-REx dataset pipeline (Day 5), this session focused on downloading, processing, and building a baseline evaluation system for **HotpotQA** — a multi-hop question-answering dataset. The goal was to construct a simple supporting-fact retrieval baseline and evaluate it using the same metrics reported in the original HotpotQA paper (Yang et al., 2018), in order to establish a reference point for future KG-based reasoning work.

Dataset Setup

- Downloaded HotpotQA via the Hugging Face *datasets* library
- **Train split:** 90,447 examples
- **Validation split:** 7,405 examples
- Verified schema: id, question, answer, type, level, supporting_facts, context
- Each example provides 10 candidate paragraphs (2 “gold” paragraphs + 8 distractors), with ground-truth supporting_facts indicating exactly which sentences justify the answer

This structure — labeled relevant vs. irrelevant evidence per question — makes HotpotQA directly useful for evaluating claim-grounding and evidence-selection methods relevant to hallucination mitigation research.

Task Definition

Supporting-fact prediction: given a question and its 10 candidate paragraphs, predict which sentences are the true supporting evidence, evaluated against the human-annotated gold supporting facts using **Precision, Recall, and F1** — the same metrics used in Table 4 of the original paper.

Experiments

Four baseline methods were implemented and evaluated on a 200-example sample from the validation set.

#	Method	Precision	Recall	F1
1	Word-overlap (raw shared words)	0.3028	0.3028	0.3028
2	TF-IDF weighted similarity (unigrams)	0.4463	0.4463	0.4463
3	TF-IDF + paragraph-title bonus (unigrams)	0.4991	0.4991	0.4991
4	TF-IDF + title bonus (bigrams)	0.4795	0.4795	0.4795

Method 1: Word-overlap baseline

Scores each sentence by counting words shared with the question, ignoring word importance. This ignores the fact that common words (e.g., “the”, “was”, “is”) are uninformative, which limits accuracy.

Method 2: TF-IDF weighted similarity

Replaced raw word-count overlap with **TF-IDF cosine similarity**, which automatically downweights common words and upweights distinctive, rare terms (e.g., entity names). This produced a substantial jump in F1 (+14.4 points over Method 1), confirming that term weighting meaningfully improves sentence relevance ranking even without any model training.

Method 3: TF-IDF + title bonus

Added a heuristic bonus when a candidate paragraph's title is explicitly mentioned in the question — a common pattern in HotpotQA's "bridge entity" questions (see Yang et al., Section 2). This gave a further improvement (+5.3 points over Method 2), reflecting that entity-mention cues carry strong signal for multi-hop questions.

Method 4: Bigram TF-IDF (paper-consistent ablation)

The original paper's own retrieval component uses **bigram TF-IDF** for document-level retrieval across 5M+ Wikipedia paragraphs (Section 5.2, Appendix C.1). Replicating this by extending the vectorizer to ngram_range=(1,2) at the **sentence level** slightly decreased performance (–2.0 points vs. Method 3).

Interpretation: This is a meaningful negative result, not an error. Bigram features are effective for coarse-grained document retrieval over very large, heterogeneous corpora (the paper's use case), but appear less effective for fine-grained sentence-level selection within a small, already-filtered set of 10 candidate paragraphs. The added vocabulary sparsity from bigrams likely diluted useful unigram signal without enough exact-phrase matches to compensate. This suggests that technique effectiveness is scale- and granularity-dependent, and should be considered when adapting retrieval methods from document-level to sentence-level tasks.

Comparison to Published Results

The original paper's trained end-to-end neural model achieves a supporting-fact F1 of **~66.6%** (distractor setting, Table 4). Our best lightweight baseline (Method 3: TF-IDF + title bonus) reaches **49.9% F1** — approximately 75% of the trained model's performance, using no training, no neural network, and only classical information-retrieval techniques.

This gap is expected and informative: it demonstrates that a meaningful portion of multi-hop supporting-fact identification can be captured with simple statistical methods, while the remaining gap likely reflects cases requiring genuine semantic understanding (e.g., paraphrased evidence with no lexical overlap) — a strong motivating case for the KG-based and embedding-based approaches central to the broader research direction (RQ1–RQ4).

Files Produced

- hotpot_train_sample.csv, hotpot_validation_sample.csv — raw processed splits
- supporting_fact_checker.py — Method 1 (word-overlap)
- supporting_fact_checker_tfidf.py — Method 2 (TF-IDF)
- supporting_fact_checker_combined.py — Method 3 (TF-IDF + title bonus)
- supporting_fact_checker_bigram.py — Method 4 (bigram ablation)

Next Steps

- Consider sentence-embedding-based similarity (e.g., sentence-transformers) as a semantic alternative to lexical TF-IDF matching
- Extend evaluation to the full validation set (7,405 examples) rather than the 200-example sample
- Connect supporting-fact selection performance to downstream answer accuracy, mirroring the paper’s “joint F1” metric (Table 4)